Roll Number:.....

Thapar Institute of Engineering & Technology, Patiala
Department of Computer Science & Engineering
Auxiliary Examination

B.E (Second Year): Semester- IV	Course Code: UCS310 Course Name: Database Management System
03 September 2025, 5.30-8.30 PM Time: 3 Hours, Max Marks:100	Name of Faculty: Dr. Geeta Kasana

Note: All questions are compulsory, and attempt all parts of a question in one place. Assume missing data, if any, suitably. Clearly specify your assumptions, if any.

Q.1 a)	Rahul is an avid reader and owns a large **personal library of books**. His colleagues and neighbours often borrow books from him, so Rahul needs a proper way to track his collection. • He maintains a list of **members**, identified by a unique **MID** (Member Identifier). With each member, Rahul records their **name** and **email address** (so he can remind them to return the books). • He also maintains a list of **books**, identified by a unique **BID** (Book Identifier). With each book, he records the **title, author,** and **genre.** • Whenever a member borrows a book, Rahul records this information along with the **date borrowed.** • When the book is returned, he notes the **date returned.** Rahul wants to maintain a complete **borrowing history** so that he can: i. Identify frequent readers. ii. Spot members who delay returning books. iii. Track which books are most popular. Draw an **E-R diagram** for this system showing: • Entities, attributes, and their identifiers. • Relationships between entities (borrows, returns). • Cardinalities (e.g., one member can borrow many books, one book can be borrowed many times over its lifetime).	10
Q1(b)	Draw a database architecture diagram and describe how DBMS modules and components interact with each other.	10
Q2 a)	Differentiate between the following with suitable examples i. Logical independence and Physical data independence ii. Lossy and lossless decomposition used in Normalization iii. Entity and Referential Integrity Constraint iv. Procedures and Trigger	20
Q3(a)	Consider a relation R (A, B, C, D, E) with FDs: AB → C, AC → B, BC → A, and D → E. i. Determine all the candidate keys of relation R. ii. Determine the given relation ***R*** is in which normal form (check up to BCNF)? Also, consider that all attributes are single-valued. Show all steps of decomposition up to BCNF.	10

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Q3(b)	What is "**Two-Phase locking protocol**"? Explain the different variants of the **Two-Phase locking protocol** with suitable examples.	10
Q4(a)	i. Discuss the ACID Properties with respect to the Transaction Management System. ii. Discuss the types of anomalies with suitable examples.	10
Q4(b)	Write the SQL statements to answer the given queries using the following relations of a Movie Rental Store ***Customers***(Customer_id, Name, City, Phone) ***Movies***(Movie_id, Title, Genre, Price) ***Rentals***(Customer_id, Movie_id, Date_rented, Date_returned) i) Find the names of customers who live in 'Delhi'. ii) List the titles of movies rented by customers from 'Mumbai'. iii) Find the IDs of customers who have never rented movies priced above Rs. 300. iv) Find the titles and genres of movies rented on '10-Jan-2021'. v) List the names of customers along with the titles of movies they have rented.	10
Q5(a)	Consider the following relation: ***Student***(rollno, sname, course_id) and ***Course***(course_id, course_name) Write the PL/SQL code to perform the following operations: i) Write a function that accepts a course id and calculates the number of students enrolled in that course and returns the count. ii) Write a procedure that accepts the course id from the already created function and returns the number of students in that course. Also, if the student count is less than 5, then delete all students enrolled in that course, followed by deleting the course itself.	10
Q5(b)	The **college administration** of *ABC University* has decided to increase the stipend of all research scholars who have not received an increment in the last two years. The stipend is increased by **15%** for such scholars. Write a PL/SQL block to update the stipend amount and display an appropriate message based on the number of rows affected by the update. Consider the following database to perform the given task: • ***Scholar*** (Scholar_ID, Name, Stipend, Last_Increment_Date, Department_ID) • ***Department*** (Department_ID, Department_Name)	5
Q5(c)	Write a cursor to calculate the total price of the first n books from the **Books** table, having columns (**bookid, booktitle, author, price**). The value of n is passed to the cursor as a parameter.	5

**********All The Best**********